In [10]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, root_mean_squared_error
```

In [11]:
```
df = pd.read_csv(r"C:\Users\q\Downloads\california_housing.csv")

# devide the data
x = df.drop("MedHouseVal",axis=1)
y= df['MedHouseVal']

#split the data into train and test
x_train, x_test, y_train, y_test = train_test_split(x,y,
                                                    test_size = 0.2,
                                                    random_state= 1234)

# create a model
lr = LinearRegression()

# train the model
lr.fit(x_train,y_train)
```

Out[11]:

▾ LinearRegression i ?

▸ Parameters

In [12]:
```
# do predictions  b0+b1*x1 + b2*x2
y_test_pred = lr.intercept_ + np.sum(lr.coef_ * x_test.values, axis = 1)
y_train_pred = lr.intercept_ + np.sum(lr.coef_ * x_train.values,axis = 1)

# calculate the error = actual - predict
df_y_train = x_train.copy()
df_y_train['y_train'] = y_train              # actual
df_y_train['y_train_pred'] = y_train_pred    # predict

# calculate error
df_y_train['error']  = y_train - y_train_pred
df_y_train['squared error'] = np.square(y_train - y_train_pred)

# calculate sse,mse,rmse
sse= np.sum(np.square(y_train - y_train_pred))
mse = np.sum(sse)/len(y_train)
rmse = np.sqrt(mse)
df_y_train['sse']= sse
```

In [13]:
```
df_y_train.tail()
```

Out[13]:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Long
8222	5.0000	21.0	3.062500	0.875000	29.0	1.812500	33.76	-1
9449	2.7530	14.0	5.789030	1.170886	1153.0	2.432489	37.39	-1
8471	4.2083	33.0	5.026163	1.151163	1073.0	3.119186	33.91	-1
17048	6.7058	44.0	6.335430	0.991614	1235.0	2.589099	37.47	-1
1318	5.4409	4.0	7.853352	1.089385	2429.0	3.392458	37.99	-1

In [14]:
```
# save the data
import joblib

d = {'model':lr,"columns":x_test.columns}
joblib.dump(d,"california2.joblib")
```

Out[14]:
```
['california2.joblib']
```

In [15]:
```
# load a model
call_model = joblib.load("california2.joblib")
call_model['model'].predict([[1,2,3,4,5,6,7,9]])
```

```
C:\Users\q\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning
: X does not have valid feature names, but LinearRegression was fitted with feature
names
  warnings.warn(
```

Out[15]:
```
array([-40.3742702])
```

In [7]:
```
import streamlit as st
```

In [ ]:
```

```

In [4]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error ,root_mean_squared_error
```

In [23]:
```
from sklearn import datasets

d = datasets.fetch_california_housing(as_frame=True).frame

x = d.drop(columns= ['MedHouseVal'])
y = d['MedHouseVal']

x_train,x_test,y_train,y_test = train_test_split(x,y,
                                                 test_size =0.2,
                                                 random_state = 52)


lrm = LinearRegression()
lrm.fit(x_train,y_train)
```

Out[23]:
```
▾ LinearRegression i ?

▸ Parameters
```

In [27]:
```
y_train_pre = lrm.intercept_ + np.sum(lrm.coef_*x_train.values,axis = 1)
y_test_pre = lrm.intercept_ + np.sum(lrm.coef_ * x_test.values, axis = 1)
```

In [33]:
```
df_train = x_train.copy()
df_train['y_train']= y_train
df_train['y_train_pre'] = y_train_pre
df_train['error'] = y_train - y_train_pre
df_train['squared error'] = np.square(y_train - y_train_pre)

sse = np.sum(np.square(y_train - y_train_pre))
mse = sse/len(y_train)
rmse = np.sqrt(mse)
mse
```

Out[33]:
```
np.float64(0.5287453554725468)
```

In [34]:
```
df_test = x_test.copy()
df_test['y_test']= y_test
df_test['y_test_pre'] = y_test_pre
df_test['error'] = y_test - y_test_pre
df_test['squared error'] = np.square(y_test - y_test_pre)

sse = np.sum(np.square(y_test - y_test_pre))
mse = sse/len(y_test)
rmse = np.sqrt(mse)
```

In [46]:
```
import joblib

d1 = {'model':lrm,"columns":x_test.columns}

joblib.dump(d1,'california_bhai_model.joblib')

train_model_1 = joblib.load('california_bhai_model.joblib')
train_model_1['model'].predict([[1,2,3,4,5,6,7,8]])
```

```
C:\Users\q\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning
: X does not have valid feature names, but LinearRegression was fitted with feature
names
  warnings.warn(
```

Out[46]:
```
array([-40.40161418])
```

In [ ]: